

MediAssist: An ML-Based Healthcare Support Tool for Diagnosis and Prescription

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Abstract—Availability of timely and correct healthcare diagnosis is still an issue, especially in resource-poor environments. Machine Learning (ML) provides promising directions to create support systems for initial diagnosis. This article presents MediAssist, a web-based application based on ML aimed at supporting users by anticipating possible diseases from their self-reported symptoms. MediAssist applies a hybrid stacking ensemble approach using Random Forest and XGBoost classifiers as the base learners and a Logistic Regression meta-learner, which is trained on an extensive symptom database encompassing 41 prevalent diseases. When inputted with symptom data through a simple human-readable interface, the trained model outputs the probable disease. Then, the application fetches and presents associated information, such as a description of the forecasted disease, precautions advised, and possible medications, from refined datasets. The hybrid model was able to obtain a test accuracy of 93.52%, which proves its potential as a useful initial healthcare support tool. This paper outlines the dataset, preprocessing methods, model design, training procedure, evaluation outcome, and the suggested web application framework.

Index Terms—Machine Learning, Disease Prediction, Symptom Analysis, Hybrid Model, Stacking Ensemble, Random Forest, XGBoost, Healthcare Support, Web Application, Diagnosis Support.

I. INTRODUCTION

Disease diagnosis in a timely manner is essential for proper treatment and patient outcomes, but problems like a lack of access to medical experts, geographical distances, and the enormity of medical knowledge continue to exist [1]. Machine Learning (ML) has been a strong resource in the healthcare industry with the potential to examine intricate data patterns and deliver meaningful insights [15]. Symptom-based disease prediction through ML, specifically, can be a useful starting point, informing people of possible health problems and pointing them in the direction of proper medical care [2].

Though traditional algorithms such as Decision Tree and random forest and many other ensemble techniques have been used for predicting disease, hybrid or ensemble models are generally more robust and efficient with enhanced performance by integrating the individual strengths of different algorithms [6]. This paper introduces MediAssist, a system designed to utilize ML towards health assistance, envisioned as a web-based application that accepts user-input symptoms [3]. MediAssist utilizes a hybrid stacking model, in this case, the combination of Random Forest (RF) and XGBoost, to forecast possible diseases. In addition to forecasting, it seeks to assist user comprehension with additional information presented, such as disease descriptions, precautions, and commonly prescribed drugs. The main contributions of this work are the creation and assessment of this RF-XGBoost stacking hybrid model for symptom-based disease prediction, the integration of prediction with additional information retrieval (description, precautions, medications), and the conceptual framework of the user-friendly MediAssist web application for deployment [2]. This document is organized to first provide an overview of prior work, secondly provide details on methodology (data, preprocessing, model architecture, evaluation), outline the implementation and system architecture under proposal, show and discuss results, provide challenges and future work, and lastly provide concluding remarks.

II. LITERATURE REVIEW

Machine learning methods have been increasingly used for disease prediction from symptoms [7]. Numerous studies have investigated and contrasted the performance of various supervised machine learning methods for this purpose. Typical methods tested include the classic classifiers Naïve Bayes and Support Vector Machines (SVM), as well as ensemble methods Random Forest. Models were typically compared by researchers to determine the best models for particular datasets or types of diseases.

For instance, research could contrast the precision, accuracy, and recall, F1-Score of Naïve Bayes, decision tree, and Random Forest on a particular symptom data set. Ensemble models such as Random Forest are often highlighted as capable of better dealing with interactions among symptoms and possibly enhancing generalization compared to simpler models. In addition, hybrid techniques, where predictions from several different models are combined (such as stacking), are being considered as one possible avenue for further improvement in prediction accuracy by tapping into the differing strengths of the underlying algorithms. Drawing from this core foundation of comparing and integrating supervised learning methods, this research applies and tests a particular hybrid stacking model that combines Random Forest and XGBoost with the goal of leveraging the advantages of both tree-based ensemble methodologies for the MediAssist use case[6].

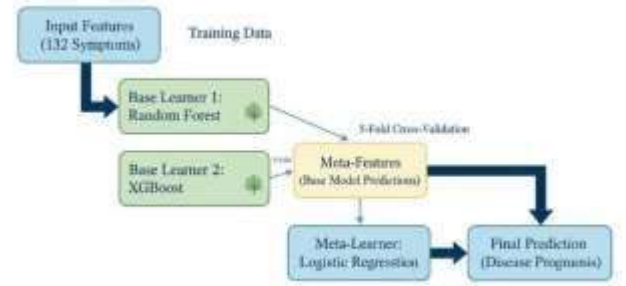
III. METHODOLOGY

A. Data Collection and Preprocessing

The research made use of four main datasets to create and contextualize the prediction model. The main Symptom Dataset (Training data.csv), which was employed for training the model, consists of 539 instances and 133 columns, of which 132 columns are separate symptoms represented binarily (1 indicating presence and 0 indicating absence) and the last 'prognosis' column reflects the identified disease, representing 41 distinct diseases [2]. Additional data came from three other datasets: the Disease Description Dataset (description.csv) translating diseases into textual descriptions, the Disease Precaution Dataset (precautions_df.csv) with recommended precautions, and the Disease Medication Dataset (medications.csv) with potential medications. Preprocessing of the symptom dataset was performed prior to training: column names were changed to lowercase with whitespace stripped; all rows with any missing values were eliminated using `dropna()`; the 'prognosis' column, being categorical, was transformed into numerical labels through Label Encoding with `sklearn.preprocessing.LabelEncoder`, retaining a mapping dictionary (diseases_list) for reverse conversion; and lastly, data was split into features (X - symptom columns) and target variable (y - encoded disease labels) [4]. Normalizing column names avoids some of the potential errors when selecting features or when manipulating the data. Removing rows that contain missing values guarantees that the machine learning algorithms have complete records, which is a requirement of many standard implementations, but can result in losing the dataset size if missing data would be common. Label Encoding the 'prognosis' column was necessary because machine learning classifiers usually work with numerical target variables instead of text labels; having a mapping enables the understanding of the model numerical output to be translated back to intelligible disease names. Lastly, explicitly keeping the features (symptoms) and the target variable (disease) apart is a basic operation in supervised learning that determines the input the model will learn from and what it tries to predict

Model Selection and Implementation

In the field of symptom-based disease prediction, several supervised machine learning algorithms are generally assessed such as Naïve Bayes, decision tree, Random Forest (RF), and Gradient Boosting such as XGBoost. Although a single model can deliver good performance, hybrid methods, especially stacking ensembles, were selected for this project with the hope of potentially performing higher accuracy and robustness. Stacking Classifier was used in implementing the hybrid model [2]. RandomForestClassifier and XGBClassifier instances were used as base learners, using certain chosen parameters suitable for the task. A LogisticRegression instance was used as the meta-learner to pool their results. StackingClassifier was configured to perform 5-fold cross-validation (`cv=5`) internally in order to train the meta-learner robustly and thus produce a strong aggregation. The entire hybrid model was then trained on the preprocessed training data (`X_train, y_train`).



B. Performance Evaluation

The performance of the constructed hybrid stacking model for disease prediction was assessed using universal, domain-specific classification metrics. Precisely, Accuracy, Precision, Recall, F1 Score were regarded as main metrics of the model's performance in identifying diseases correctly from symptoms. Accuracy gave a measure of the overall correctness of all predictions for all classes [7]. Precision and Recall were determined through the use of a macro average to determine the performance of the model as a whole on all 41 different disease classes, reflecting the model's correctness in making positive predictions and recognizing all true cases, respectively. In addition, a Confusion Matrix was also created to provide a class-by-class, detailed visualization of the model's predictive accuracy, particularly pointing out the areas of strength or confusion.

C. Ethical Considerations

The creation and utilization of the MediAssist system followed rigorous ethical behavior, specifically relating to the use of data and application of model [14]. Symptom dataset for training the healthcare prediction model were drawn from publicly available or simulated records, presumed de-identified for ensuring patient privacy.

For the proposed web application, any user symptom data collection would require strict compliance with data privacy regulations and secure handling procedures. . The principal worry is that the system, though, will always be extremely open about what it can and cannot do; the software will consistently be explicitly identified as a support tool, not as a replacement for diagnostics. The User Interface will have Disclaimers that will make the user aware that he/she has to get in touch with licensed medical practitioners. Besides, the issue of trust in the AI's ability to maintain the equilibrium and to handle the biases that might arise from the training data is another factor that has a significant impact on its responsible use [12]. The character of the data and the dedication to ethical standards were the main factors that determined the design process which defined when the AI would be safe and appropriate. The basic architectures of the Random Forest and XGBoost algorithms are illustrated in Figures 5 and 6 respectively.

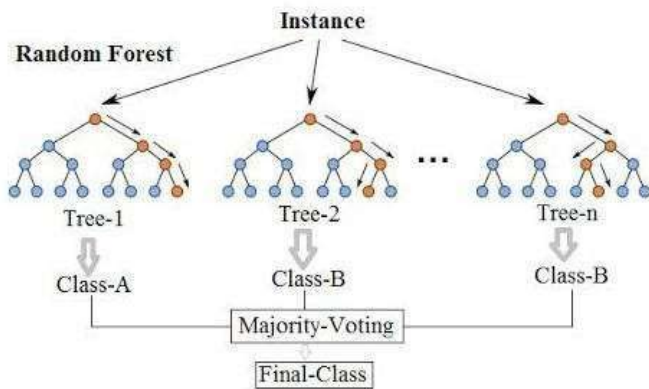


Fig. 2. Random forest Architecture.

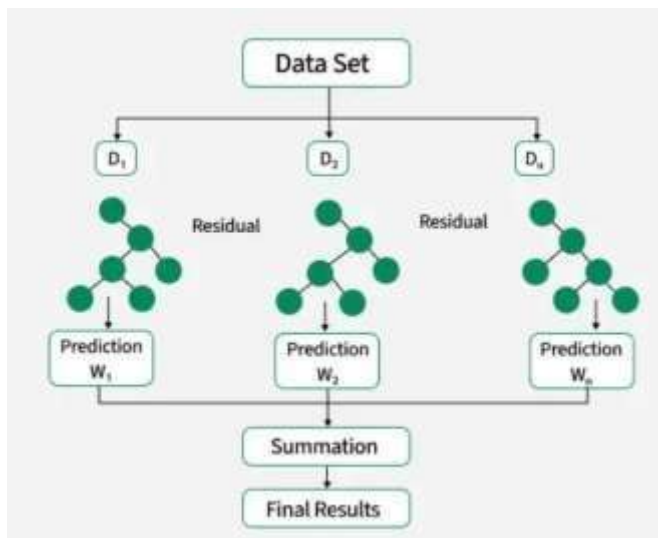


Fig. 3. XGBoost Architecture

Data Confidentiality and Privacy:

Goal: Each and every symptom data which the user presents is guaranteed of privacy, and the health-related matters, which are a result of the model's prediction but not associated with the user's symptoms, are kept secret.

Method: An assumed de-identified or simulated training set was used for the purpose of the research. The public interface would have to implement strict data management practices (secure transfer like HTTPS, very limited storage, adherence to privacy laws). User data will be treated in a secure manner.

Challenges The right approach to privacy matters and compliance with data protection regulations are some of the difficulties in product development that need users' voices and a just point of view.

Accuracy, User Safety, and Scope Limitation:

Goal: The project aims at reducing the likelihood of the model being misused as a diagnostic tool to a certain extent and covering the overall model's accuracy limitations partly.

Method: The application interface displays unique and explicit disclaimers (refer to Figure 4). These disclaimers unambiguously point out that MediAssist is just a backup tool, gives suggestions on the basis of statistical trends, and does not replace a doctor's professional diagnosis. Users are strongly warned to get in touch with licensed healthcare professionals. The information given is just for informational purposes.

Challenges: User expectations must be met with correct management and the user must be assumed to go through all necessary disclaimers printed in small letters and even aspire to comprehend them..

Bias and Fairness:

goal: The reason for this is to ensure that all user groups perform equally and also to prevent any biases that might be hidden in the training data from becoming more pronounced.

Approach: The standard preprocessing took place and the performance was evaluated in a macro-average manner so as to include all classes. It is recommended to perform consistent bias monitoring through fairness metrics for every version released as part of the continuous improvement process.

Challenges: A machine learning-based approach has to deal with the quality of data that is used for the training and testing of models, one of the main if not the main issue. Among other problems the researchers have to solve daily are: properly assessing and disclosing biases in health-related data, collecting huge data sets that are so diverse that they can be used for both training and testing, and perhaps, sacrificing overall accuracy for the sake of fairness among different demographic groups.

Transparency and Interpretability:

Goal: It is fostered by fostering trust between the user and model when interpreting the predictions.

Approach: The currently applied ensemble model is very much recognized for its limited inherent interpretability. In addition, future research may consider using different model interpretation approaches such as SHAP and LIME to show the symptoms that have the most impact on the prediction process.

Challenges: Given the different technical skill levels of users, their understanding of the interaction sounds' interpretability may differ significantly between expert audiences and less expert audiences.

Informed Consent and User Control:

Objective When the users are made aware of the purpose, the data, and the tool's limitations, it becomes easier for them to understand the main reason behind their willingness to submit sensitive data.

Approach: It is recommended to have a plain statement at the outset of the user engagement (refer to Figure 4) that would make explicit the aim of the instrument, which is not diagnostic, how the information will be treated, and what privacy measures will be implemented, and at the same time, obtain the user's clear consent to continue.

Challenges: Setting up accountable pathways, ones that are clear and transparent, should always stay as the main goal.

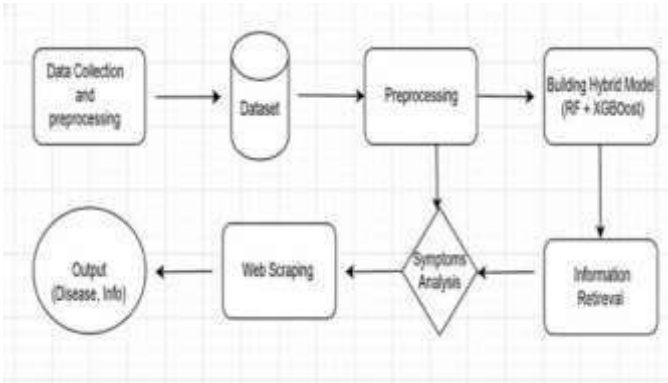


Fig. 4. Flow Analysis.

IV. IMPLEMENTATION

The MediAssist system is specifically created as an available web-based application meant to act as an initial reference for persons who would like to learn about possible health conditions related to their symptoms [3]. Its main aim is to offer preliminary disease diagnosis recommendations and related healthcare information, filling the gap between symptom occurrences and consulting medical professionals. The system does so through the integration of three essential elements: an easy-to-use symptom entry interface, a strong machine learning algorithm (the hybrid RF-XGBoost stacker) for prediction, and specially prepared supplementary datasets for background information [15]. The integration not only creates efficiency in processing user requests but also furnishes a holistic, though initial, result within the accessibility of an online portal.

A. Machine Learning for Disease Prediction

Providing predictions about diseases via the symptoms indicated is the main function of MediAssist. The application operates on the basis that the user first selects the symptoms. The selected symptoms form a binary vector that is aligned with the 132 features used in the training process. The features are then input to the pre-trained hybrid stacking model that integrates Random Forest, XGBoost, and Logistic Regression as a meta-learner, which is the model referred to in the [1] citation. The model, which has been trained on the preprocessed Training data.csv dataset, then assigns one disease out of 41 possible ones as the most likely outcome. Metrics such as accuracy, precision, and recall were used to validate the reliability of the model during the testing phase. After a disease has been predicted, the system changes the numerical label to its corresponding name and retrieves related information, such as disease definitions,

precautions, and probable medications, from the extra datasets (description.csv, precautions_df.csv, medications.csv). The presentation format gives the patients not only a possible diagnosis but also instant, relevant context and advice. The presentation format gives the patients a possible diagnosis along with instant, relevant context and advice. The primary procedure of the users using the application MediAssist is identification of symptoms. Fast the user's picks are changed to 132-dimensional binary vector representation. This satisfies the request of the system to do the processing. Next, the binary vector is dispatched to the hybrid_model.pkl file which is the next step. The prediction of the three models Random Forest (RF), XGBoost, and Logistic Regression afterwards gets a lot of attention. Instantly, the number assigned is converted to the respective disease name via the diseases_list dictionary. The predicted name acts as a key to get the extra data sets. Therefore, it shows the associated description, precautions, and medication lists [4]. The application's highlight is the gradual transition from symptom input to prediction and then information retrieval. Users receive the contextual data which is directly related to the model's outputs.

Web Application Interface

MediAssist runs as a web application. It keeps a straightforward and systematic appearance overall. The main aim stays focused on simplifying things for users seeking basic health information quickly. The interface divides into distinct stages. This setup guides individuals through the process in a logical process:

Login Page: First and foremost, the login page is the first thing to show up. It is, however, a very simple authentication system that could turn out to be very helpful as the users would later on be able to return to their previous symptom logs or even obtain a history of their predictions. Besides, they might also have access to their personal preferences. Nevertheless, the entire issue of data privacy would have to be thoroughly considered first.

Symptom Input Page: Initially and throughout, the symptom input page serves as the primary point of user interaction. The model's 132 symptoms are presented in such a way that their very pleasant design invites even the most hesitant users. The users can make their selection either through checkboxes or a multi-select dropdown menu. This method is really nice and really simple to point out the symptoms without facing any problem. Each symptom has to be provided with the most unambiguous and simple description. After the user inputs have been done, the back end is responsible for everything. It handles processing and then the prediction is done.

Output page: After the backend has completed its processing, the output page is displayed. It presents the results in a highly structured and easy-to-scan format. The predicted disease name is prominently highlighted in large font at the very beginning. Afterwards, the information is partitioned into sections. One section features a short description that is taken from the description.csv file..

Recommended Precautions: Displayed perhaps in a bullet point format for readability, derived from precautions_df.csv.

Potential Medications: The list was compiled solely for informational purposes, with medications.csv being the source. Nevertheless, this page would reiterate the disclaimer stating that the information is not intended to replace professional medical advice text.

V. CHALLENGES AND FUTURE ENHANCEMENTS

Even with its solid start at 93.52 percent accuracy, the MediAssist system still faces a few key limitations. The main issue comes from the dataset's limited size and range, just 539 examples covering 41 diseases. That setup hampers how well it generalizes, so it really needs a lot more diverse data from real clinical sources down the line. Adding that kind of growth would boost its strength and cut down on any built-in biases. Before thinking about real-world use, experts like qualified doctors have to validate it thoroughly. On top of that, the user interface should keep clear disclaimers in place. Those would stress that MediAssist serves as a helpful tool, not a replacement for proper diagnosis, all to keep users safe [16]. Future tweaks ought to tackle things like making the model more interpretable. They should also handle richer symptom details, say on severity or how long symptoms last. Plus, setting up regular updates for supporting medical info makes sense. In the end, rolling this out as a secure web app that scales well and stays accessible means fixing big hurdles in security, user experience, and privacy for data.

VI. RESULT

The real-world importance of these findings is that the MediAssist system shows great potential as an initial healthcare assistance tool. Its ability to suggest possible diseases based solely on self-reported symptoms may be especially beneficial in underserved or rural areas where medical experts are not readily available. By providing low-cost, data-driven early advice and relevant information, such as disease descriptions, precautions, and medications, MediAssist could help patients better understand their symptoms and make informed decisions about their medical care. While the confusion matrix below offers detailed insights into the specific strengths and weaknesses of predictions, the overall positive results support the app's function as an effective first source of information, highlighting the need for further consultation with medical professionals.

TABLE I. Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	95.67%	0.9577	0.9604	0.9590
XGBoost	88.89%	0.8614	0.8508	0.8420
Decision Tree	67.90%	0.7758	0.7280	0.7450
Hybrid (RF + XGBoost)	93.52%	0.9492	0.9138	0.9080

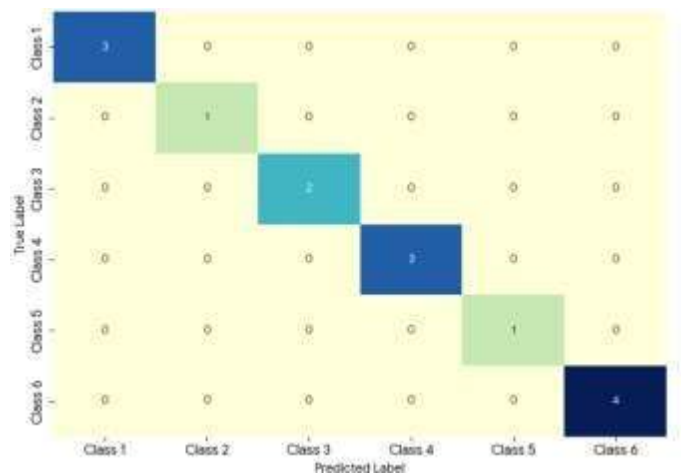


Fig. 5. Hybrid Model – Confusion Matrix.



Fig. 6 Web page Interface.

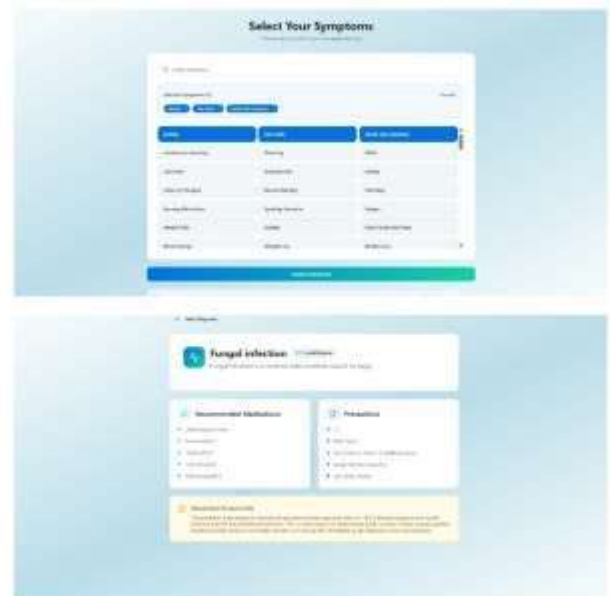


Fig.7 Prediction of disease

VII. CONCLUSION

In this Paper we explore the MediAssist system employs a hybrid Random Forest and XGBoost stacking model for initial symptom-based disease prediction with robust predictive performance. Through integration of predictions with available disease descriptions, precautions, and drugs information, it has promise to act as an effective initial healthcare supporting tool, particularly in resource-poor settings. Notably, MediAssist is an informational tool, and not a substitute for medical professional diagnosis.

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